Fantasy League Score Prediction Using LightGBM Gradient-Boosted Trees

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ABSTRACT

Fantasy League Score Prediction Using LightGBM Gradient-Boosted Trees

Fantasy sports have gained significant popularity, creating a need for data-driven methods to predict player performance and fantasy scores. This project, **"Fantasy League Score Prediction Using LightGBM Gradient-Boosted Trees,"** aims to develop a machine learning model that predicts the fantasy points a player is likely to score in an upcoming match based on historical performance and match-related features. The system uses player statistics such as recent fantasy points, batting or scoring performance, bowling or defensive performance, playing time, player form, opponent strength, home/away conditions, and other relevant factors.

The collected data is preprocessed to handle missing values, remove inconsistencies, and transform the required features. Feature engineering is then performed to generate meaningful indicators such as recent average points, performance trends, and player form. The **LightGBM gradient-boosting algorithm** is used to train the prediction model because of its efficiency, high predictive performance, and ability to handle complex relationships between multiple features. The model is evaluated using metrics such as **Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score**.

The proposed system can help fantasy sports users make more informed player-selection decisions by providing predicted fantasy scores and player rankings. The project demonstrates how machine learning and historical sports data can be combined to develop an intelligent fantasy score prediction system.

Signature of the Guide